KEY

17 — **Prime number**
A blue box indicates that the number is prime. It has no factors other than 1 and itself.

42 (2 3 7) — **Composite number**
A yellow box denotes a composite number, which means that it is divisible by more than 1 and itself.

smaller numbers show whether the number is divisible by 2, 3, 5, or 7, or a combination of them

6 (2 3)	7	8 (2)	9 (3)	10 (2 5)
16 (2)	17	18 (2 3)	19	20 (2 5)
26 (2)	27 (3)	28 (2 7)	29	30 (2 3 5)
36 (2 3)	37	38 (2)	39 (3)	40 (2 5)
46 (2)	47	48 (2 3)	49 (7)	50 (2 5)
56 (2 7)	57 (3)	58 (2)	59	60 (2 3 5)
66 (2 3)	67	68 (2)	69 (3)	70 (2 5 7)
76 (2)	77 (7)	78 (2 3)	79	80 (2 5)
86 (2)	87 (3)	88 (2)	89	90 (2 3 5)
96 (2 3)	97	98 (2 7)	99 (3)	100 (2 5)

Prime factors

Every number is either a prime or the result of multiplying together prime numbers. Prime factorization is the process of breaking down a composite number into the prime numbers that it is made up of. These are known as its prime factors.

$$30 = \underbrace{5}_{\text{prime factor}} \times \underbrace{6}_{\text{remaining factor}}$$

To find the prime factors of 30, find the largest prime number that divides into 30, which is 5. The remaining factor is 6 (5 x 6 = 30), which needs to be broken down into prime numbers.

$$6 = \underbrace{3}_{\text{largest prime factor}} \times 2$$

Next, take the remaining factor and find the largest prime number that divides into it, and any smaller prime numbers. In this case, the prime numbers that divide into 6 are 3 and 2.

list prime factors in descending order

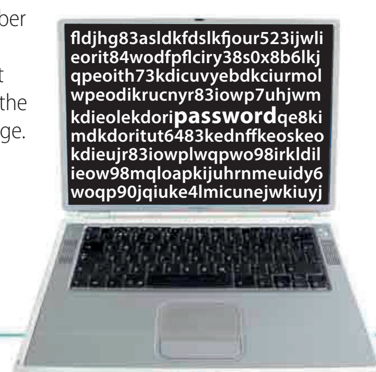
$$30 = 5 \times 3 \times 2$$

It is now possible to see that 30 is the product of multiplying together the prime numbers 5, 3, and 2. Therefore, the prime factors of 30 are 5, 3, and, 2.

REAL WORLD

Encryption

Many transactions in banks and stores rely on the Internet and other communications systems. To protect the information, it is coded using a number that is the product of just two huge primes. The security relies on the fact that no "eavesdropper" can factorize the number because its factors are so large.



▷ **Data protection**
To provide constant security, mathematicians relentlessly hunt for ever bigger primes.